

Introduction

This work presents a method for analyzing the 3D geometry of individual fibers in high-resolution volumetric datasets. The analysis has been done using a python notebook and the whole code can be found under <https://github.com/LukasEgkher22/FiberDetection>. The approach includes:

- **2D Detection:** Segment fibers in each (x,y) slice
- **3D Tracking:** Reconstruct fibers as continuous 3D curves across the z-direction
- **Orientation Analysis:** Measure azimuth and elevation angles of each fiber
- **Curvature Measurement:** Quantify fiber bending along their trajectories
- **Statistical Evaluation:** Generate histograms of orientation and curvature
- **Resolution Comparison:** Assess method performance on downsampled (low-resolution) data

Samples

UD: Unidirectional glass fiber composite
MOCK: Air-textured glass fiber composite

Acquisition

Zeiss Xradia 520 Versa XCT
Voxel size: 1.989 μm
Field of view: 2 mm

Fiber detection in each slice - (x,y) directions

We segment individual fibers by first **applying a Gaussian filter** to each 2D slice, reducing noise and highlighting fiber-like structures. **Peak detection** is then performed on the filtered image to identify local maxima corresponding to fiber centers. This process is repeated **slice by slice** across the 3D volume, with parameter tuning for both UD and MOCK materials. Finally, **visual overlays** of the detected centers help assess the segmentation results in both full and zoomed-in views.

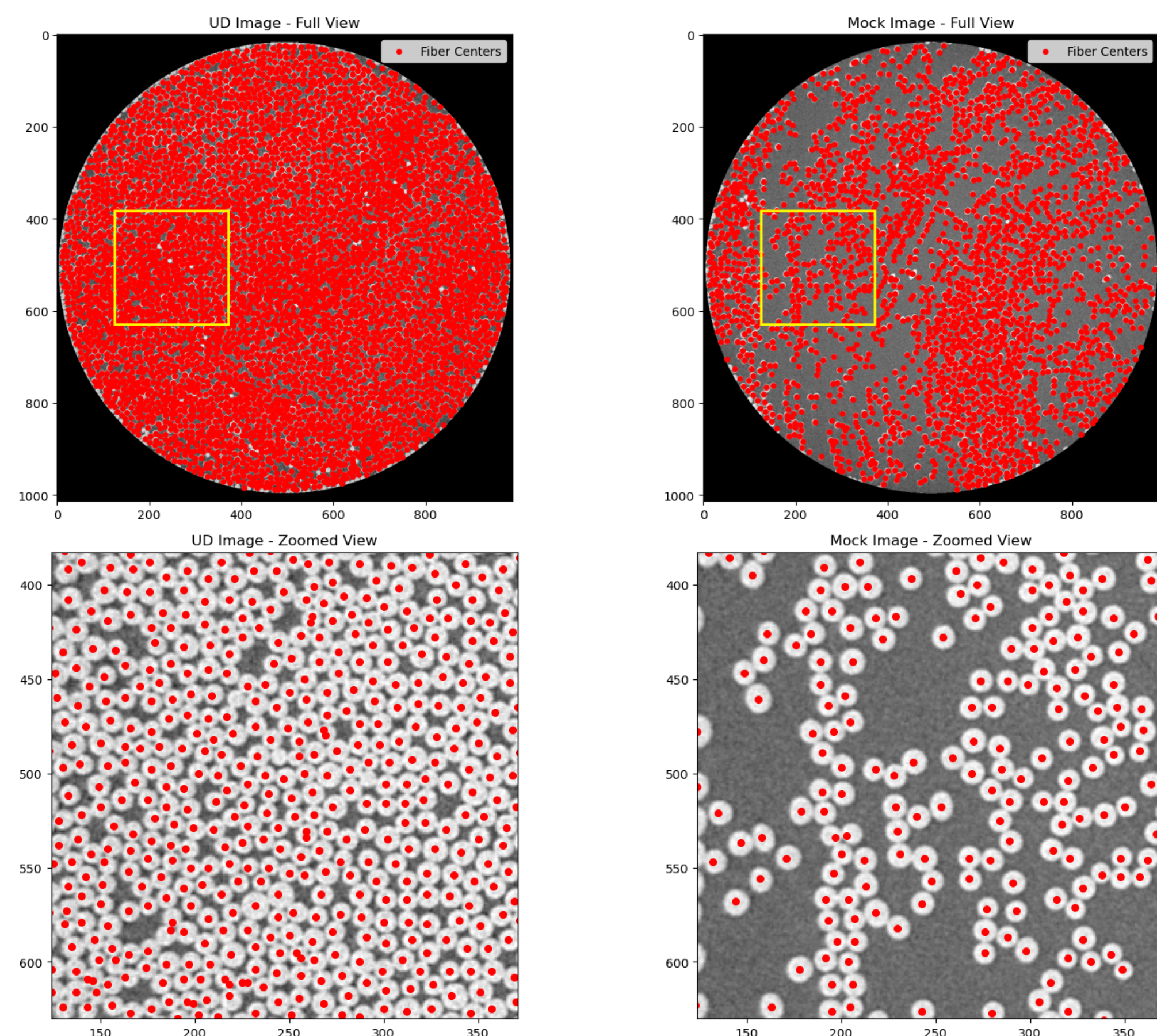


Figure 1. Fiber detection in UD and MOCK samples with zoomed-in views.

Track fibers in z-direction as 3D curves

Fiber centers detected in each slice are linked into 3D tracks using a method adapted from *Anders Bjorholm Dahl's* [1] open-source trackpoints algorithm. This algorithm connects points slice-by-slice based on **spatial proximity** and **continuity**. Tracks shorter than a predefined minimum length are **filtered** out to improve the reliability of subsequent orientation and curvature analysis. The resulting 3D curves provide a geometric representation of individual fibers throughout the volume.

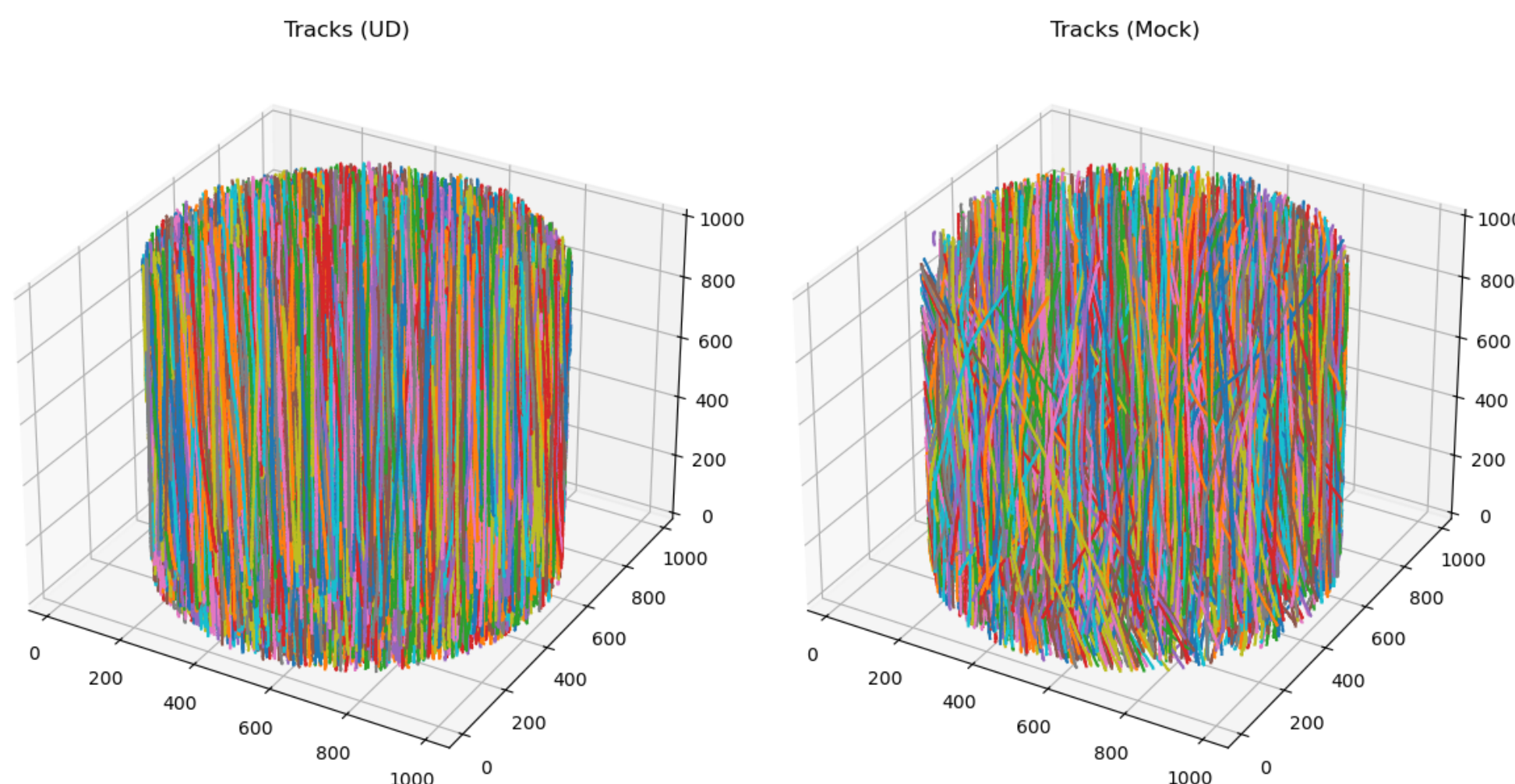


Figure 2. 3D fiber tracks reconstructed from slice-wise center detections in UD and MOCK volumes.

Angular definitions

Azimuth angle: Represents the angle in the xy-plane (0° to 360°)

- 0° = positive x-axis direction
- 90° = positive y-axis direction
- 180° = negative x-axis direction
- 270° = negative y-axis direction

Elevation angle: Represents the angle from the xy-plane (-90° to 90°)

- 0° = fiber lies flat in the xy-plane
- 90° = fiber points straight up (positive z-axis)
- -90° = fiber points straight down (negative z-axis)

Analysis of fiber orientation

UD fibers

- UD fibers show a very high elevation angle (**average: 88°**), which underlines that most of the fibers are **aligned vertically with the z-axis**.
- The azimuth angle for UD fibers is not as relevant due to their strong z-axis alignment. However, it appears to be more or less evenly distributed, with an **average of 208.82°** , local (maxima) peaks at 0° , 90° , 180° and 270° .

Mock fibers

- Mock fibers exhibit a more random distribution of elevation angles, indicating less alignment with the z-axis. **The average elevation angle is 79.4°** .
- The azimuth angle is more relevant for mock fibers. **It shows a peak around 130°** , and the majority of azimuth angles lie between 90° and 180° , as visible in the bottom-right plot.

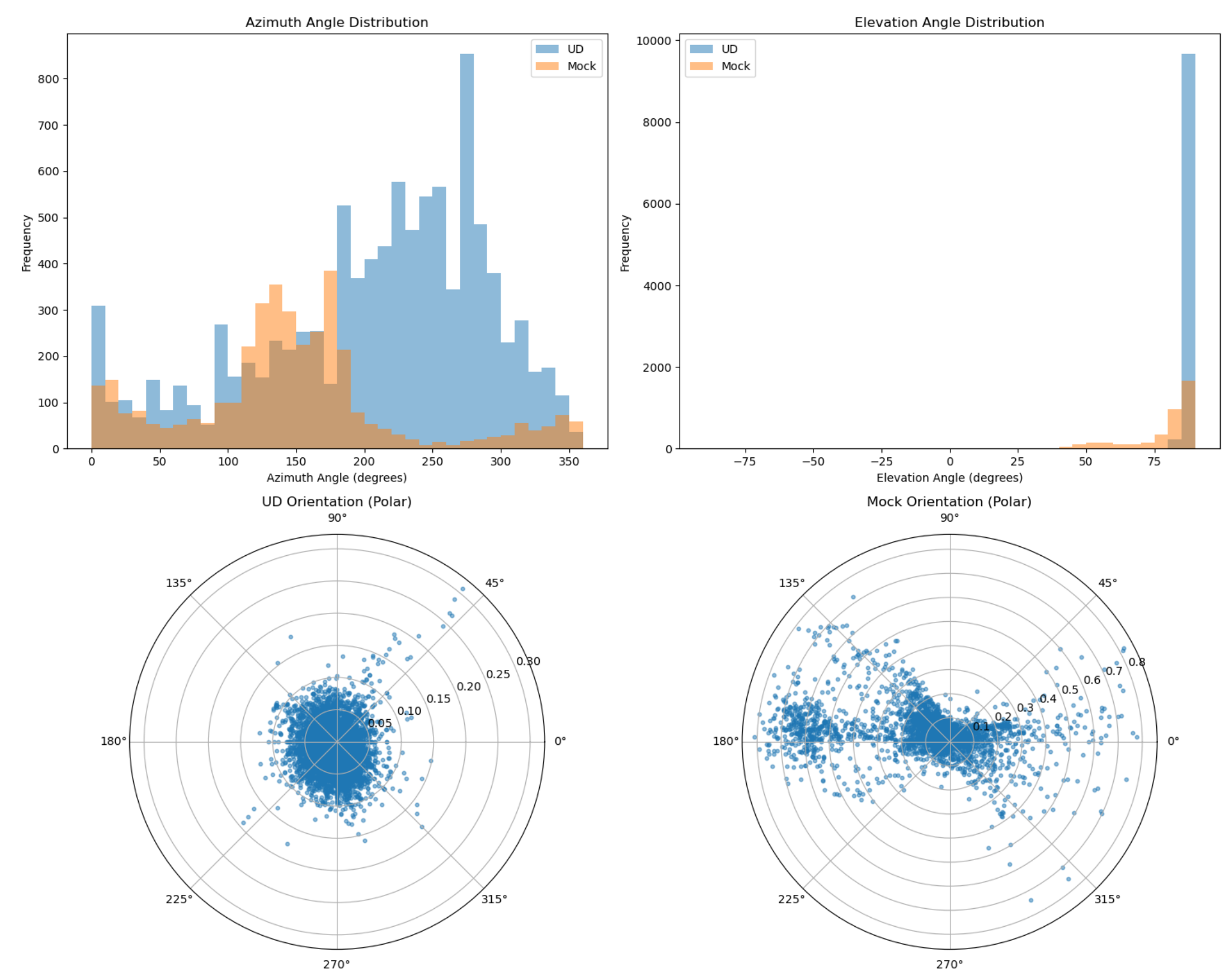


Figure 3. Azimuth and elevation angle distribution in fiber orientation

Analysis of fiber curvature

- The curvature of UD fibers ranges from **0.3 to 1.4** for tracks with small lengths and then gets more and more constant around **0.8**.
- The curvature of mock fibers ranges more at the beginning as well. To be specific it ranges from **0.3 to 1.2** for short tracks and get a little closer until the maximum length (1000) where it ranges from **0.4 to 0.9**.

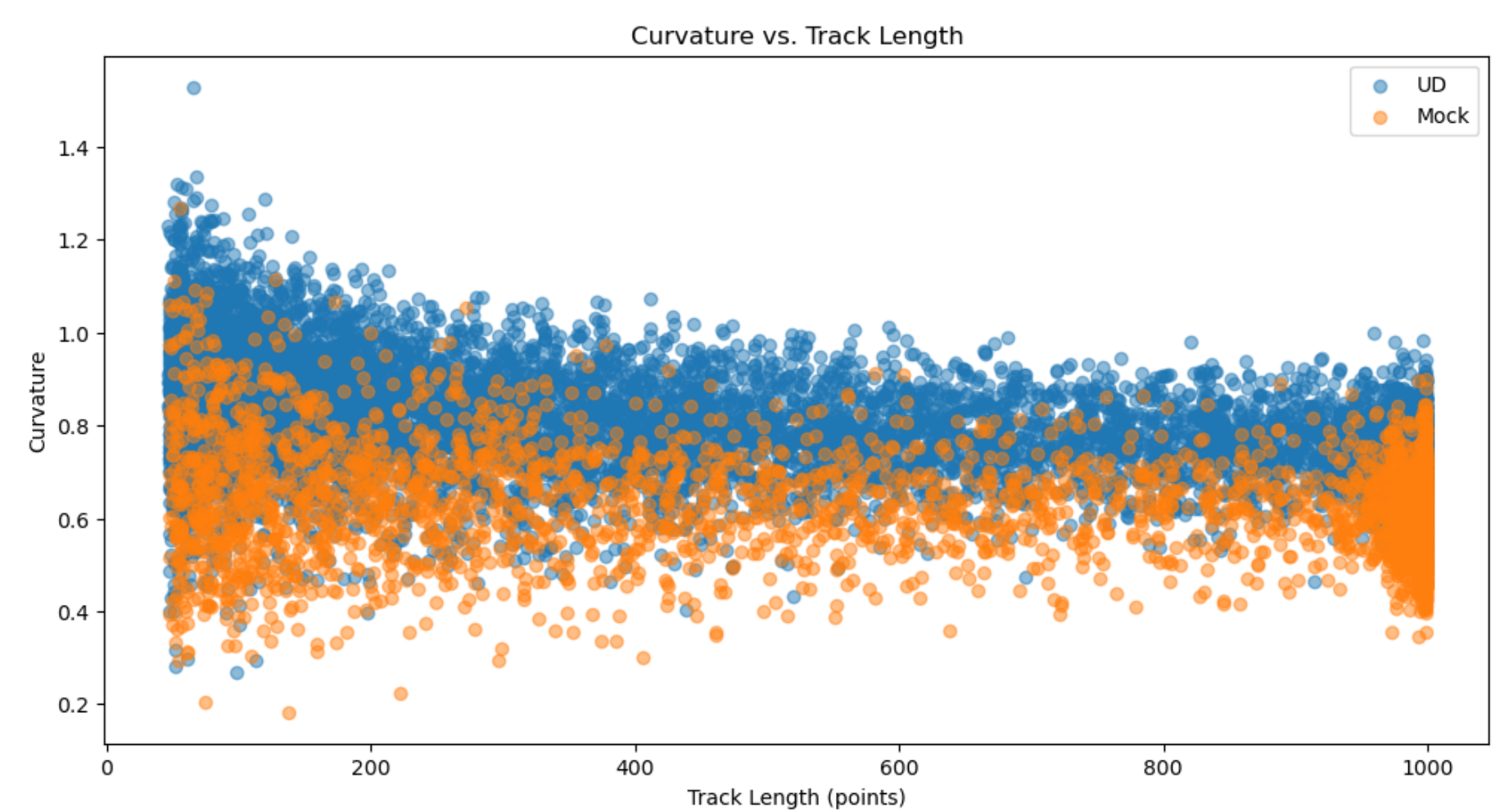


Figure 4. Fiber curvature distribution based on frequency in UD and Mock data

References

[1] Anders Bjorholm Dahl.
<https://github.com/abdahl/3dtools>.
Accessed: 2025-05-03.